

MOHAMED ZAMEER Z

Software Development Engineer | Python | FastAPI | React | AI/ML
+91 7338021017 | zameer.trichy@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Development Engineer with hands-on experience building production backend and full-stack applications using Python, FastAPI, React, PostgreSQL, Redis, and Docker. Experienced in REST API development, microservices, ERP and e-commerce platforms, fraud monitoring, banking reconciliation, OCR, and machine learning. Promoted from AI/ML Developer Intern to Associate Software Development Engineer based on delivery across multiple production systems.

PROFESSIONAL EXPERIENCE

Innovitegra Solutions

Associate Software Development Engineer | Jun 2026 - Present

AI/ML Developer Intern | Jan 2026 - Jun 2026

Selected Projects

Enterprise ERP Platform — Backend & Full Stack

- Architected the backend of a production multi-module ERP system handling HR, attendance, CTMS/staff augmentation, billing, invoicing, email management, CRM, permissions, and notifications
- Designed and implemented the CTMS attendance and billing workflow including multi-day attendance tracking, draft/publish state management, approval flows, and backend-driven invoice generation
- Integrated transactional email delivery with template versioning, safe HTML rendering via Jinja2, and production debugging through Hostinger Mail API
- Maintained stable REST API contracts and Swagger documentation supporting React frontend integration across employee management, client onboarding, and issue tracking modules

E-Commerce Marketplace Platform — Backend & Full Stack

- Engineered the backend of a 3-portal marketplace (Merchant, Admin, Customer) covering catalog management, product approval workflows, pricing engine, cart, orders, returns, and AI-powered product quality review via Ollama
- Developed the complete Admin Portal frontend in React, including approval flows, AI quality reports, merchant management, order tracking, dashboards, and notifications

Banking Transaction Reconciliation Platform — Frontend

- Developed the complete frontend for a banking transaction reconciliation platform enabling comparison of ATM and ACH transaction files to detect and resolve mismatches
- Designed reconciliation dashboards and data views supporting real-time filtering, review, and tracking of mismatched transactions across multiple file types

Multilingual Identity Document OCR System — ML Engineering

- Fine-tuned a PaddleOCR recognition model on GPU for Pashto and Dari text extraction from identity documents, using a pretrained Arabic recognition model as the base
- Engineered a synthetic data augmentation pipeline using RTL text rendering, motion blur, shadow, compression, and perspective transforms to improve model robustness on real-world document images

Fraud Risk Monitoring Platform — Backend

- Developed a real-time fraud risk monitoring system with automated risk scoring, Redis-backed processing, and PostgreSQL for high-volume transaction storage and analysis
- Deployed scalable microservices via Docker with structured logging and rate limiting to support production traffic with sub-second response times

PROJECTS

AI Resume Screener | *Python, NLP, TF-IDF, XGBoost, SHAP*

- Developed an NLP-based resume screening system using TF-IDF and cosine similarity to match and rank candidate resumes against job descriptions
- Integrated XGBoost for candidate classification and SHAP for explainability, surfacing skill gaps and generating actionable improvement recommendations

AI Chest X-Ray Pneumonia Detection | *PyTorch, DenseNet121, Grad-CAM, Tesseract OCR, Streamlit*

- Developed a deep learning pipeline using DenseNet121 to classify chest X-rays for pneumonia detection, improving accuracy through augmentation and fine-tuning
- Integrated Grad-CAM for visual explainability and Tesseract OCR for patient metadata extraction; deployed end-to-end via Streamlit

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, SQLAlchemy, Pydantic, Flask, REST APIs, Swagger, Jinja2

Frontend: React, API Integration, Responsive Design

Databases: PostgreSQL, MySQL, Redis, ClickHouse

AI and ML: PyTorch, Scikit-learn, XGBoost, NLP, TF-IDF, PaddleOCR, OpenCV, SHAP

DevOps: Docker, Git, GitHub, Linux

EDUCATION

B.E. Computer Science and Engineering (AI and ML)

Sai Vidya Institute of Technology | CGPA: 8.6 | 2022 - 2026

2nd PUC

Chethana PU College | 2020 - 2022

CERTIFICATIONS

- NPTEL — Deep Learning (IIT Ropar) | IBM — AI Fundamentals
- Certisured — EDA for Machine Learning | Analogica — Machine Learning Training

ACHIEVEMENTS

- Promoted from AI/ML Developer Intern to Associate Software Development Engineer at Innovitegra Solutions
- Maintained 8.6 CGPA in B.E. AI and ML | Active Member, Computer Society of India (CSI)